

AEGIS

AUTONOMOUS ENTERPRISE GROWTH INTELLIGENCE
SYSTEM

Technical Requirements Document

TRD v1.0

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Author	AEGIS Engineering Team

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1. Introduction & Scope

1.1 Purpose

This Technical Requirements Document (TRD) translates the AEGIS Product Requirements Document (PRD) into concrete technical specifications, architecture decisions, and engineering guidelines. It serves as the authoritative engineering reference for all development teams building AEGIS.

1.2 Scope

AEGIS is an AI-powered enterprise operating system that combines Agentic AI, Generative AI, Hybrid RAG, Knowledge Graphs, Long-Term Memory, and Business Intelligence into a unified platform. This TRD covers all 25 core modules defined in the PRD, their technical implementation, integration points, and non-functional requirements.

1.3 Definitions & Acronyms

Term	Definition
AEGIS	Autonomous Enterprise Growth Intelligence System
RAG	Retrieval-Augmented Generation
RBAC	Role-Based Access Control
MFA	Multi-Factor Authentication
KG	Knowledge Graph
ToT	Tree-of-Thought (reasoning strategy)
BM25	Best Matching 25 – ranking function for text retrieval
BGE	BAAI General Embedding model for semantic search

2. System Architecture Overview

2.1 High-Level Architecture

AEGIS follows a **modular microservices architecture** with an event-driven backbone. The system is decomposed into five architectural tiers:

PRESENTATION TIER		
Next.js App · React Components · Tailwind CSS · TS		
API GATEWAY		
FastAPI · JWT Auth · Rate Limiting · CORS		
AGENT TIER	INTELLIGENCE	DATA PROCESSING TIER
LangGraph	TIER	Celery Workers
CrewAI	Hybrid RAG	Redis Queue
AutoGen	Knowledge Graph	Document Parsers
Orchestrator	Long-Term Memory	Embedding Pipeline
DATA TIER		
PostgreSQL · Supabase · Qdrant · ChromaDB · Redis		
OBSERVABILITY TIER		
LangSmith · OpenTelemetry · Prometheus · Grafana		

2.2 Deployment Architecture

The system is designed for cloud-native deployment using containerized services orchestrated via Kubernetes (K8s). Each module is an independently deployable unit communicating via REST APIs and Redis Pub/Sub for async events.

Component	Deployment	Scaling Strategy
Frontend (Next.js)	Vercel / CDN Edge	Edge-based auto-scale
API Gateway (FastAPI)	K8s Pods	Horizontal Pod Autoscaler (HPA)
Agent Workers	K8s Pods (GPU-enabled)	Queue-depth based scaling
Celery Workers	K8s Pods	Task-queue based scaling

PostgreSQL	Managed (Supabase / RDS)	Read replicas
Qdrant	Managed / Self-hosted cluster	Shard-based horizontal
Redis	Managed (ElastiCache)	Cluster mode

3. Technology Stack Specifications

3.1 Frontend Stack

Technology	Version	Purpose	Justification
Next.js	14.x+	Full-stack React framework	SSR, API routes, app router, edge functions
React	18.x+	UI component library	Component model, concurrent features, suspense
TypeScript	5.x+	Type-safe development	Compile-time error detection, IDE support
Tailwind CSS	3.x+	Utility-first styling	Rapid UI development, design consistency
Zustand / Jotai	Latest	State management	Lightweight, minimal boilerplate
React Query	5.x+	Server-state management	Caching, background sync, optimistic updates
Socket.IO Client	4.x+	Real-time communication	WebSocket with fallback transport

3.2 Backend Stack

Technology	Version	Purpose	Justification
FastAPI	0.110+	API framework	Async native, OpenAPI auto-docs, Pydantic validation
Python	3.11+	Backend language	AI/ML ecosystem, async/await, type hints
Celery	5.x+	Task queue	Distributed task processing, scheduling

Redis	7.x+	Cache / broker / pub-sub	In-memory speed, Celery broker, session store
Pydantic	2.x+	Data validation	Schema enforcement, serialization
SQLAlchemy	2.x+	ORM	Async support, relationship mapping
Alembic	1.x+	DB migrations	Version-controlled schema evolution

3.3 AI/ML Stack

Technology	Purpose	Details
LangGraph	Agent orchestration	Stateful multi-step agent workflows with cycles and branching
CrewAI	Multi-agent collaboration	Role-based agent teams with delegation capabilities
AutoGen	Conversational agents	Multi-agent conversation patterns, code execution
Qdrant	Vector database	High-performance similarity search, filtering, payload support
ChromaDB	Lightweight vector store	Development/prototyping vector store, in-process embedding
BM25 (Rank-BM25)	Keyword retrieval	Sparse retrieval for hybrid search alongside semantic vectors
BGE Embeddings	Text embedding	BAAI bge-large-en-v1.5 for high-quality semantic embeddings
Neo4j / NetworkX	Knowledge Graph	Entity-relationship modeling, graph traversal queries

3.4 Data & Storage Stack

Technology	Purpose	Data Types
PostgreSQL 15+	Primary relational DB	Users, orgs, CRM, financial records, audit logs
Supabase	Backend-as-a-Service layer	Auth, real-time subscriptions, storage, edge functions

Redis 7+	Cache, sessions, queues	Session tokens, rate-limit counters, task queues, pub/sub
Qdrant	Vector storage	Document embeddings, semantic search indices
S3 / MinIO	Object storage	Uploaded files, generated documents, media assets

3.5 Monitoring & Observability

Tool	Purpose	Metrics
LangSmith	LLM observability	Token usage, latency, chain traces, prompt debugging
OpenTelemetry	Distributed tracing	Request traces, span timing, error propagation
Prometheus	Metrics collection	API latency, error rates, queue depth, CPU/memory
Grafana	Dashboards & alerting	Visualization of all metrics, alert rules, SLA tracking

4. Module Technical Requirements

4.1 Authentication & Authorization (Module 1)

Requirement	Specification	Priority
OAuth 2.0 / OIDC	Google, GitHub, Microsoft SSO via Supabase Auth	P1
MFA	TOTP-based (Google Authenticator), SMS fallback	P1
RBAC	7 roles (Founder, Sales Mgr, Marketing Mgr, HR, Finance, Support, Admin) with granular permissions matrix	P1
Organization Multi-tenancy	Tenant isolation at DB level (schema-per-tenant or row-level security)	P1
Session Management	JWT access tokens (15 min) + refresh tokens (7 days) stored in Redis	P1
API Key Management	Scoped API keys for integrations, rate-limited per key	P2

4.2 Organization Setup & Data Ingestion (Module 2)

Requirement	Specification	Priority
PDF Ingestion	PyMuPDF / pdfplumber for parsing, chunking (512 tokens, 50 token overlap)	P1
CRM Import	CSV/Excel upload, Salesforce/HubSpot API connectors	P2
Website Crawling	Scrapy / BeautifulSoup with depth-limited crawling, sitemap support	P2
Database Connectors	PostgreSQL, MySQL, MongoDB read adapters via SQLAlchemy	P3
Embedding Pipeline	BGE embeddings → chunking → Qdrant upsert via Celery workers	P1

4.3 AI Workspace (Module 3)

Requirement	Specification	Priority
Business Chat	WebSocket-based streaming responses, markdown rendering, code blocks	P1
Voice Input	Web Speech API + Whisper API fallback, real-time transcription	P2
File Uploads	Drag-and-drop, max 50MB, supported: PDF, DOCX, XLSX, CSV, images	P1
Conversation Memory	Per-session context window + long-term memory retrieval from KG	P1
Source Citations	Inline source attribution with clickable references to source documents	P1

4.4 Multi-Agent AI System (Modules 4–5)

Agent	Role	Tools / Capabilities
CEO Agent	Strategic oversight, task delegation	All sub-agent delegation, report aggregation, scenario planning
Sales Agent	Pipeline, forecasting, lead scoring	CRM read/write, email drafting, meeting scheduling

Marketing Agent	Campaign analysis, content strategy	Analytics APIs, content generation, A/B test analysis
Finance Agent	Budgeting, forecasting, compliance	Financial data queries, report generation, anomaly detection
HR Agent	Recruitment, policy queries	Job posting, resume screening, policy RAG
Legal Agent	Contract review, compliance	Document analysis, clause extraction, risk flagging
Research Agent	Market research, competitive intel	Web search, data aggregation, trend analysis
Analytics Agent	Data analysis, visualization	SQL generation, chart creation, statistical analysis

Agent Orchestration Pattern: LangGraph manages the primary agent state machine. CrewAI handles multi-agent collaboration for complex tasks. AutoGen provides conversational patterns for human-in-the-loop workflows. All agents share a common tool registry and memory layer.

4.5 Hybrid RAG System (Module 6)

Component	Specification
Semantic Search	Qdrant vector similarity (cosine), top-k=20, with metadata filtering
Keyword Search	BM25 ranking over tokenized corpus, top-k=20
Fusion Strategy	Reciprocal Rank Fusion (RRF) combining semantic + keyword results
Re-ranking	Cross-encoder re-ranking (bge-reranker-large) on fused results, top-k=5
Knowledge Graph	Graph-based entity retrieval for structured relationships, merged into context
Chunking Strategy	Recursive character splitter, 512 tokens, 50 token overlap, metadata preservation
Embedding Model	BAAI/bge-large-en-v1.5 (1024 dims), batch embedding via GPU workers

4.6 Knowledge Graph (Module 7)

Entity extraction from ingested documents using NER models. Relationship mapping stored in Neo4j. Supports traversal queries for contextual retrieval (e.g., "What products does Client X use?")

→ entity hop from Client → Products).

4.7 Long-Term Memory (Module 8)

Persistent memory stored as vector embeddings + structured summaries. Enables agents to recall past conversations, decisions, and organizational context. Memory types: episodic (conversations), semantic (facts), procedural (workflows).

4.8 Business Digital Twin (Module 9)

A live simulation model of the organization built from real-time data feeds. Supports scenario planning ("What if we increase marketing spend by 20%?"), predictive modeling, and what-if analysis using Monte Carlo simulations.

4.9 Business Intelligence Dashboards (Module 10)

Requirement	Specification
Chart Library	Recharts / Tremor for React-native charts
Real-time Updates	WebSocket push for live KPI changes
Custom Dashboards	Drag-and-drop widget builder, saved layouts per user
Export	PDF, PNG, CSV export of dashboards and individual charts
AI Insights	Auto-generated narrative summaries of dashboard data via LLM

4.10 Intelligence Modules (Modules 11–17)

Module	Key Technical Requirements
AI Lead Generation	Web scraping pipeline, lead scoring ML model, CRM integration API
CRM Intelligence	Deal prediction model, customer health scoring, churn detection
Marketing Intelligence	Campaign ROI calculator, sentiment analysis, content performance API
Financial Intelligence	Cash flow forecasting, anomaly detection, budget variance analysis
Customer Intelligence	NPS analysis, ticket categorization, support metrics dashboard

Meeting Intelligence	Whisper transcription, action item extraction, sentiment tracking
Document Intelligence	OCR pipeline, auto-classification, key field extraction, version control

4.11 Automation & Generation (Modules 18–20)

Module	Specification
Workflow Automation	Visual workflow builder (node-based), trigger-action model, Celery execution
Proposal Generator	Template engine + LLM drafting, PDF output, version history
Presentation Generator	Slide-by-slide LLM generation, PPTX export via python-pptx

4.12 Platform Modules (Modules 21–25)

Module	Specification
Notifications	In-app (WebSocket), email (SendGrid), push (Firebase FCM), webhook
Integrations	REST API connectors: Slack, Teams, Gmail, Calendar, Zapier webhooks
Admin Panel	User management, org settings, usage analytics, billing (Stripe)
AI Model Hub	Model registry, A/B testing, cost-aware routing (GPT-4 vs Claude vs local)
Monitoring	LangSmith traces, OpenTelemetry spans, Prometheus metrics, Grafana dashboards

5. API & Integration Specifications

5.1 API Design Principles

- RESTful API with OpenAPI 3.1 specification (auto-generated by FastAPI)
- Versioned endpoints: /api/v1/...
- Consistent error responses: { "error": { "code": "...", "message": "...", "details": [] } }
- Pagination: Cursor-based for lists, limit + cursor parameters
- Rate limiting: Token bucket algorithm, per-user and per-API-key limits

5.2 Key API Endpoints

Endpoint	Method	Description
/api/v1/auth/login	POST	Email/password login, returns JWT
/api/v1/auth/oauth/{provider}	GET	OAuth redirect for SSO
/api/v1/org/{id}/ingest	POST	Upload documents for RAG ingestion
/api/v1/chat/stream	WebSocket	Streaming AI chat responses
/api/v1/agents/{agent}/invoke	POST	Invoke specific agent with task
/api/v1/search	POST	Hybrid RAG search query
/api/v1/knowledge-graph/query	POST	Knowledge graph traversal
/api/v1/dashboards	GET/POST	CRUD for BI dashboards
/api/v1/workflows	GET/POST	CRUD for workflow automation

5.3 WebSocket Channels

Channel	Purpose	Payload
chat:{session_id}	AI chat streaming	Token-by-token LLM response
notifications:{user_id}	Real-time alerts	Notification objects
dashboard:{dashboard_id}	Live dashboard updates	Metric deltas
agents:{task_id}	Agent progress tracking	Status updates, intermediate results

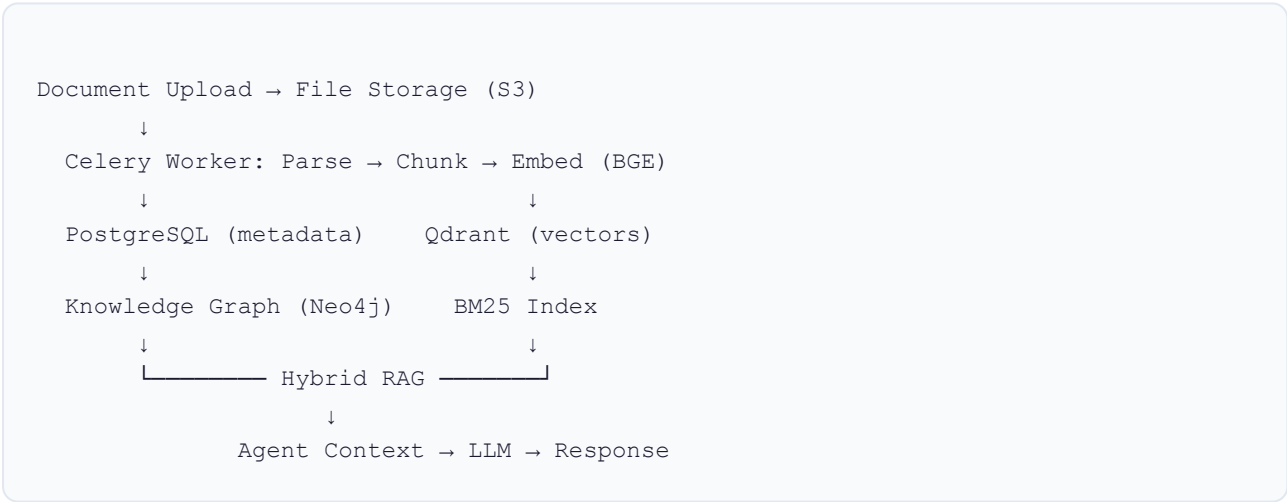
6. Data Architecture

6.1 Database Schema (Core Entities)

Table	Key Columns	Relationships
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organizations	id, name, plan, settings, created_at	Has many: users, documents, agents
users	id, org_id, email, role, mfa_enabled	Belongs to: organization
conversations	id, user_id, agent_type, messages[], created_at	Belongs to: user
documents	id, org_id, filename, type, status, chunks_count	Has many: chunks
document_chunks	id, doc_id, content, embedding_id, metadata	Belongs to: document
agent_tasks	id, agent_type, status, input, output, tokens_used	Belongs to: conversation
workflows	id, org_id, name, trigger, actions[], status	Belongs to: organization
memory_entries	id, org_id, type, content, embedding_id, timestamp	Belongs to: organization

6.2 Data Flow



7. Security Requirements

Category	Requirement	Implementation
Encryption at Rest	All data encrypted	AES-256 for DB, S3 server-side encryption
Encryption in Transit	TLS 1.3 everywhere	Certificate management via Let's Encrypt / ACM

Authentication	Multi-factor required for admin roles	TOTP + SMS via Supabase Auth
Authorization	Row-level security	PostgreSQL RLS policies + application-level RBAC
Input Validation	All inputs sanitized	Pydantic schemas, SQL parameterization, XSS prevention
Prompt Injection	LLM input guarding	Input/output guardrails, prompt sandboxing, content filtering
Audit Logging	All mutations logged	Append-only audit_log table with user, action, timestamp, diff
Secret Management	No hardcoded secrets	HashiCorp Vault / AWS Secrets Manager, .env for dev
GDPR/SOC2	Compliance framework	Data retention policies, right-to-deletion, access logs

8. Performance & Scalability Requirements

Metric	Target	Measurement
API Response Time (P95)	< 200ms	Prometheus histogram
Chat First Token Latency	< 1.5s	LangSmith trace
RAG Retrieval Latency	< 500ms	Qdrant + BM25 combined
Document Ingestion	100 pages/min per worker	Celery task metrics
Concurrent Users	1,000+ per org	Load test (k6/Locust)
System Uptime	99.9% SLA	Uptime monitoring
Vector Search (1M docs)	< 100ms	Qdrant benchmark
Agent Task Completion	< 30s for simple tasks	LangSmith span timing

Scalability Note: The system must support horizontal scaling at every tier. No single points of failure.

All stateful services (DB, cache, vector store) must be clustered or managed with automatic failover.

9. Infrastructure & DevOps

9.1 CI/CD Pipeline

- **Source Control:** Git (GitHub/GitLab), trunk-based development
- **CI:** GitHub Actions – lint, type-check, unit tests, integration tests on every PR
- **CD:** Auto-deploy to staging on merge to develop, manual promotion to production
- **Container Registry:** Docker Hub / GitHub Container Registry
- **Infrastructure as Code:** Terraform for cloud resources, Helm charts for K8s

9.2 Environment Strategy

Environment	Purpose	Data
Development	Local development	Docker Compose, seeded test data
Staging	Pre-production validation	Anonymized production-like data
Production	Live system	Real data, full monitoring

10. Testing Strategy

Test Type	Tools	Coverage Target
Unit Tests	pytest (backend), Jest (frontend)	≥ 80% code coverage
Integration Tests	pytest + TestContainers	All API endpoints, DB operations
E2E Tests	Playwright / Cypress	Critical user journeys
Load Tests	k6 / Locust	1000+ concurrent users
AI/RAG Tests	RAGAS, custom eval suite	Retrieval accuracy ≥ 85%, answer relevance ≥ 80%
Security Tests	OWASP ZAP, Bandit	No critical/high vulnerabilities

11. Phased Delivery Plan

Phase 1 – Foundation (Weeks 1–8)

Deliverable	Modules	Dependencies
Auth System	Module 1	Supabase setup, PostgreSQL
AI Chat Interface	Module 3	FastAPI, WebSocket, LLM API keys
Hybrid RAG Pipeline	Modules 2, 6	Qdrant, BGE model, Celery
Core Agents (CEO, Sales)	Modules 4, 5	LangGraph, tool registry
Basic Dashboard	Module 10	React charts, API endpoints

Phase 2 – Intelligence (Weeks 9–18)

Deliverable	Modules	Dependencies
CRM Intelligence	Module 12	CRM connectors, ML models
Marketing Intelligence	Module 13	Analytics APIs, content analysis
Financial Intelligence	Module 14	Financial data models
Meeting Intelligence	Module 16	Whisper API, action extraction
Knowledge Graph	Module 7	Neo4j, NER models
All Remaining Agents	Modules 4, 5	Phase 1 agent framework

Phase 3 – Enterprise (Weeks 19–28)

Deliverable	Modules	Dependencies
Business Digital Twin	Module 9	All data pipelines, simulation engine
Autonomous Planning	Advanced Features	Tree-of-Thought, scenario planning
Enterprise SaaS Features	Modules 21–25	Billing, admin, SSO, compliance

Workflow Automation	Module 18	Visual builder, Celery orchestration
Document/Presentation Gen	Modules 19, 20	Template engine, PPTX library

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